

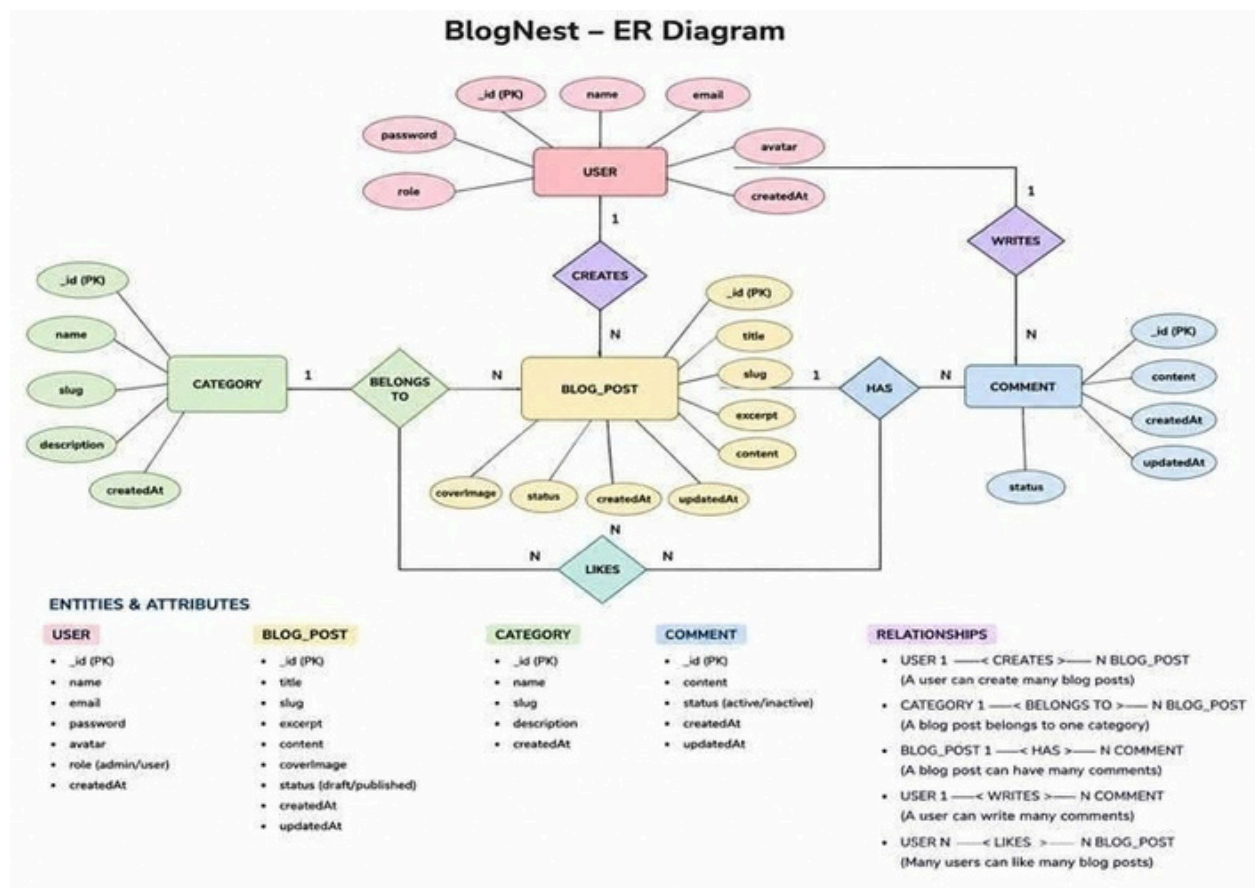
Phase 4

Project Planning Phase

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Entity-Relationship (ER) Diagram Description

The underlying document structure manages several collections bound by logical application references:



1. User Entity

- **_id** (Primary Key - ObjectId)
- **name** (String, Required)
- **email** (String, Required, Unique)
- **password** (String, Hashed, Required)
- **role** (String, Default: "reader")

2. Post Entity

- **_id** (Primary Key - ObjectId)
- **userID** (Foreign Key referencing User Collection)
- **name** (String, Required - represents Title)
- **photo** (Object/String pathway link)
- **message** (String - Core content body)
- **likes** (Number, Default: 0)
- **timestamps** (True)

Key Structural Mappings

- **User to Post (One-to-Many):** An author can submit multiple blog documents.
- **User to Comment (One-to-Many):** Readers/Users can append multiple response comments across independent post nodes.
- **Post to Comment (One-to-Many):** A single blog document stores individual referenced comment submissions.

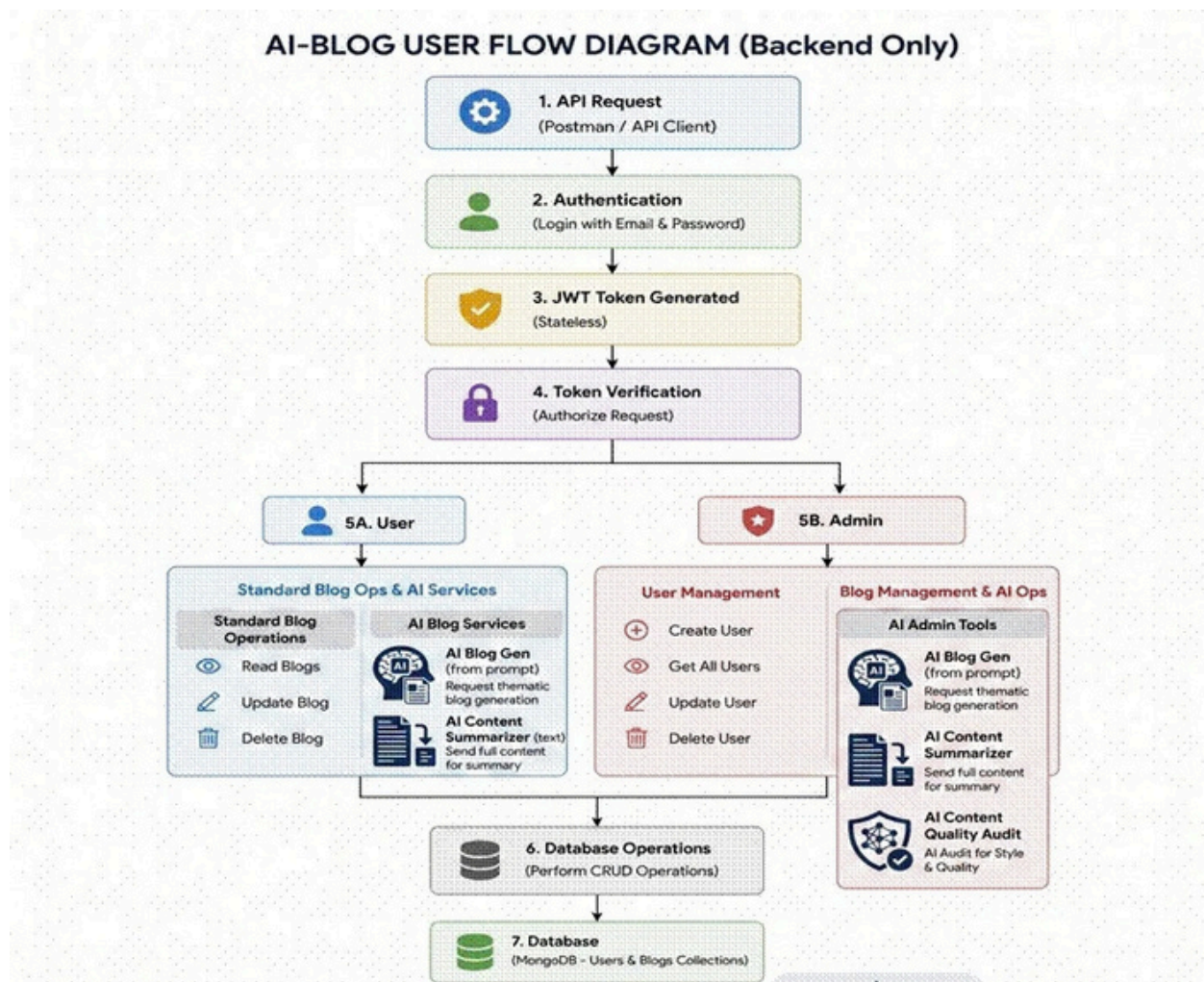
Key Features

- **Role-Based Route Guarding:** Strict token processing blocks unauthorized tiers from modifying core settings.
- **CRUD Blog Execution Engine:** Complete schema management over content definitions, status states, and asset assignment paths.
- **Categorization System:** Native tag indices facilitate filtering and querying for enhanced discoverability.
- **Comment Moderation Logic:** Protected entry points for updating moderation flags (approved, pending, spam).
- **Sanitization Safeguards:** Payload parameter sanitation prevents XSS vectors and NoSQL parameter manipulation injections.
- **I-Driven Blog Generation Pipeline:** Integrates external Large Language Model (LLM) abstractions and context-aware prompt templates to synthesize full-length, structured blog posts from high-level thematic inputs.
- **Intelligent Content Summarization Engine:** Leverages semantic compression layers to parse extensive article payloads, dynamically generating concise abstracts and automated metadata summaries.

Roles and Responsibilities

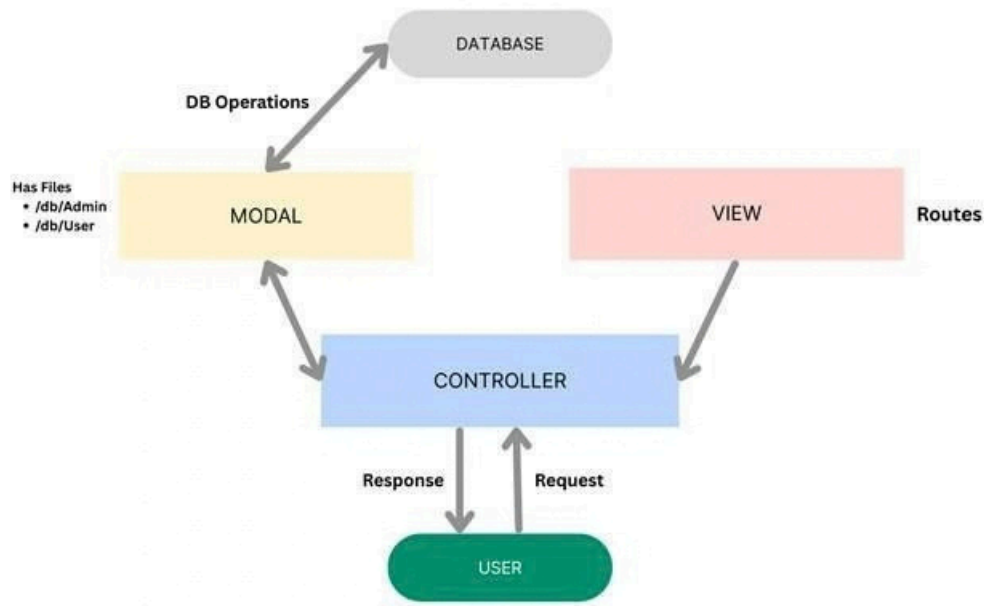
- **Admin:** Complete environment override control. Can manage user database registrations, alter role assignments, prune malicious comments, and inspect full logging entries.
- **Editor:** Focuses on editorial workflows. Reviews pending draft posts, marks records for instant publication, and handles category cleanup scripts.
- **Author:** Restricted account access. Can issue creation statements only for posts bound to their explicit account record.
- **Reader:** Consumer access profile. Reads published materials and posts comments under an ongoing authenticated identity.

User Flow:



MVC Architecture Pattern

The system architecture structures its source configuration layers around the Model-View-Controller design pattern to completely decouple components:



Model Layer (Data Blueprint)

Defines structure blueprints via strongly enforced Mongoose schemas mapping explicit fields into raw collection items inside MongoDB. It sets default states, indexing rules, and transformation parameters before validation persistence occurs.

Controller Layer (Execution Matrix)

The intermediary brain of the ecosystem. It captures request vectors from the execution routes, checks structural request constraints, delegates processing workflows directly to models, and packages the resultant output into standard JSON return arrays.

View Layer (Routing Network Interface)

In this pure headless backend service context, standard user-interface template views do not exist. Instead, the interface layer lives as an API **Routing Layer**. It links network requests on specified paths directly to their associated controller handler logic maps.